1 Random Variables

2 Probability distributions

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Motivating Example

Consider a tossing of two coin experiment,

S = {HH,HT,TH,TT}, X(number of heads) = {0,1,2}. Then

$$P(X = 0) = P(\{TT\}) = 1/4$$

$$P(X = 1) = P(\{HT, TH\}) = 2/4 = 1/2$$

$$P(X = 2) = P(\{HH\}) = 1/4$$

Here, X is called **random variable**.

Definition

A *random variable* is a function that assigns a real number to each sample point in the sample space of a random experiment.
It is denoted by X. That is,

$$X : S \to \mathbb{R}.$$

Another Example: Sum of Two Dice

When two fair dice are tossed, the sample space consists of ordered pairs

$$S = \{(i, j) : i, j = 1, 2, \ldots, 6\}.$$

Let X denote the sum of the two dice. For example,

$$P(X = 2) = P(\{(1, 1)\}) = \frac{1}{36}$$

$$P(X = 3) = P(\{(1, 2), (2, 1)\}) = \frac{2}{36}$$

$$P(X = 4) = P(\{(1, 3), (2, 2), (3, 1)\}) = \frac{3}{36}$$

Although the outcomes differ, they all produce the same value of the random variable $X = x$.

Note: $X = x$ means the random variable takes the value x.

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Probability distribution

The set of all possible values of random variable X, along with their corresponding probabilities is called its probability distribution.

Example: Let X be the random variable defined as the sum of the values obtained when two fair dice are thrown. The probability distribution of X is given in the following table:

x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Discrete Random Variable

A random variable X is called a *discrete random variable* if it can take only a finite or countably infinite number of distinct values. The probability distribution of a discrete random variable is given by

$$P(X = x_i) = p(x_i), \quad i = 1, 2, 3, \ldots$$

Probability Mass Function

Let X be a discrete random variable. The function $p(x)$ defined by

$$P(X = x_i) = p(x_i), \quad i = 1, 2, 3, \ldots$$

is called the *probability mass function* (PMF) of X.
The PMF satisfies the following properties:

1. $p(x_i) \geq 0 \quad$ for all i
2. $\sum_i p(x_i) = 1$

Cumulative Distribution Function

Let X be a discrete random variable

$$\text{c.d.f.} \qquad F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

Note: For a discrete random variable X, $F(x)$ satisfies

Properties

1. $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$
2. $0 \leq F(x) \leq 1$
3. If $x \leq y$ then $F(x) \leq F(y)$

Example

Let X be the random variable defined as the sum of the values obtained when two fair dice are thrown.

x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Note that

$$(i)\, p(x_i) \geq 0, \quad (ii) \sum_{x=2}^{12} P(X=x) = 1 \qquad \textbf{PMF}$$

$$(i) F(6) = P(X \leq 6) = 1/36 + 2/36 + 3/36 + 4/36 + 5/36 \qquad \textbf{CDF}$$

$$(ii)\, 0 \leq F(x) \leq 1 \quad \textit{for all } x \qquad (iii)\, \textit{If } 2 \leq 3 \textit{ then } F(2) \leq F(3)$$

Observe: Need to verify all possible values.

Mean or Expected value

What is Expected value?

If X is a discrete random variable that can take values

$$x_1, x_2, x_3, \ldots$$

then the expected value of X, written as μ or $E[X]$, is defined as

$$\mu = E[X] = \sum_i x_i \, P(X = x_i)$$

(i.e.) "If I repeat the experiment many times, what will be the average value of the outcomes?"

What is Variance?

Variance measures how spread out the values of X are around the expected value. It is defined using expectation:

$$\sigma^2 = \mathrm{Var}(X) = E\left[(X-\mu)^2\right] = E(X^2) - (E(X))^2$$

(i.e.) How far do values move away from the Expected value?

What is Standard deviation?

Standard deviation is just the square root of variance:

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

(i.e.) How much does X typically differ from its expected value?

Two Examples - Comparison

Student A - coming time	Student B -coming time
Problem Details: $X =$ travel time (minutes) Values: $28, 29, 30, 31, 32$ $P(X = x) = \frac{1}{5}$ each	**Problem Details:** $Y =$ travel time (minutes) Values: $10, 20, 30, 40, 50$ $P(Y = y) = \frac{1}{5}$ each
Expected Value: $E[X] = \sum xP(X = x)$ $= \frac{1}{5}(28 + 29 + 30 + 31 + 32)$ $= 30$	**Expected Value:** $E[Y] = \sum yP(Y = y)$ $= \frac{1}{5}(10 + 20 + 30 + 40 + 50)$ $= 30$
Variance: $\mathrm{Var}(X) = \sum(x - 30)^2 P(X = x)$ $= \frac{1}{5}(4 + 1 + 0 + 1 + 4)$ $= 2$	**Variance:** $\mathrm{Var}(Y) = \sum(y - 30)^2 P(Y = y)$ $= \frac{1}{5}(400 + 100 + 0 + 100 + 400)$ $= 200$
Standard Deviation: $\sigma_X = \sqrt{\mathrm{Var}(X)}$ $= \sqrt{2} \approx 1.41$	**Standard Deviation:** $\sigma_Y = \sqrt{\mathrm{Var}(Y)}$ $= \sqrt{200} \approx 14.14$

Expected Value of a Function of a Discrete Random Variable

Let X be a discrete random variable with probability mass function $p(x)$. Then the expected value of a function $h(X)$ is defined as

$$E[h(X)] = \sum_x h(x)p(x)$$

Result 1

If a and b are fixed numbers, then

$$E(aX + b) = aE(X) + b$$

Result 2

If a and b are fixed numbers, then

$$V(aX + b) = a^2 V(X)$$

Problems

Check whether the following functions are probability mass functions (PMFs)

1. $p(x) = \frac{1}{4}, \quad x = 1, 2, 3, 4$
2. $p(x) = \frac{x-1}{6}, \quad x = 0, 1, 2, 3$
3. $p(x) = \frac{x}{6}, \quad x = 1, 2, 3$
4. $p(x) = \begin{cases} 0.2, & x = 0 \\ 0.5, & x = 1 \\ 0.3, & x = 2 \end{cases}$
5. $p(x) = \begin{cases} 0.6, & x = 1 \\ -0.1, & x = 2 \\ 0.5, & x = 3 \end{cases}$

Problems

Convert the following probability mass functions (PMFs) into cumulative distribution functions (CDFs):

1. $p(x) = \begin{cases} 0.2, & x = 0 \\ 0.5, & x = 1 \\ 0.3, & x = 2 \end{cases}$
2. $p(x) = \frac{1}{4},\ x = 1, 2, 3, 4$
3. $p(x) = \frac{x}{6},\ x = 1, 2, 3$
4. $p(x) = \begin{cases} 0.1, & x = 1 \\ 0.2, & x = 2 \\ 0.3, & x = 3 \\ 0.4, & x = 4 \end{cases}$
5. $p(x) = \frac{1}{2^x},\ x = 1, 2, 3, \ldots$

Problems

A die is tossed three times. A success is defined as obtaining a 1 or 6. Let X denote the number of successes in the three tosses. Find the expected value of successes.

Solution

$$P(\text{success}) = P(1 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}, \quad P(\text{failure}) = \frac{2}{3}.$$

The random variable X can take the values $0, 1, 2, 3$.

x	0	1	2	3
$P(X = x)$	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$

The expected value of X is

$$E[X] = \sum xP(X = x).$$

$$E[X] = 0 \cdot \frac{8}{27} + 1 \cdot \frac{12}{27} + 2 \cdot \frac{6}{27} + 3 \cdot \frac{1}{27} = 1.$$

Problems

Two fair dice are thrown. Let $X = \max\{\text{First dice value, Second die value}\}$. Find the probability distribution of X and its mean.

Solution

The random variable X can take the values

$$X = 1, 2, 3, 4, 5, 6.$$

Probability distribution of X is given by

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$E[X] = \sum xP(X = x).$$

$$E[X] = \frac{1}{36}(1 \cdot 1 + 2 \cdot 3 + 3 \cdot 5 + 4 \cdot 7 + 5 \cdot 9 + 6 \cdot 11) = \frac{161}{36}.$$

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Bernoulli Distribution

A random variable X is said to follow a Bernoulli distribution if it takes only two possible values:

$$X = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1-p, \end{cases} \qquad 0 \le p \le 1.$$

Expected Value

$$E[X] = \sum x\, P(X = x).$$

Since X takes only the values 0 and 1,

$$E[X] = 1 \cdot P(X = 1) + 0 \cdot P(X = 0).$$

$$E[X] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

Variance

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2.$$

Since $X^2 = X$ for $X = 0, 1$,

$$E[X^2] = E[X] = p.$$

Therefore,

$$\mathrm{Var}(X) = p - p^2 = p(1-p).$$

The probability mass function (PMF) of X is defined as

$$P(X = x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0. \end{cases}$$

Equivalently, the PMF can be written in compact form as

$$P(X = x) = p^x (1-p)^{1-x}, \quad x = 0, 1.$$

Binomial Random Variable

A random variable X that counts the number of successes in many independent, identical, Bernoulli trials.

Note:

1. **Independent:** The outcome of any trial is independent of the outcome of other trials.
2. **Identical:** The probability of success in each trial must be constant.
3. **Bernoulli trial:** Each trial results in either success or failure.

Example:

Toss a coin many times.

Illustration of a situation:

Toss a coin 5 times. Let the success be getting "Head".

Let the probability of success $= 0.6$. Say, we need 3 successes out of 5 trials.

There are $5C_3$ ways to get this 3 successes out of 5 trials. Each way has probability $(0.6)^3(0.4)^2$.

Observe

	0.6	0.6	0.6	0.4	0.4
1st *way*	S	S	S	F	F
	0.6	0.4	0.4	0.6	0.6
2nd *way*	S	F	F	S	S
.					
.					
.					
$5C_3$ *ways*					

Binomial distribution

Let a random experiment consists of n independent, identical, Bernoulli trials. Let p denotes probability of success. Let $X =$ Number of trials that result in success.

$$\boxed{X \sim B(n,p) \quad \textit{iff} \quad P(X=x) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x},\ x = 0,1,2,\ldots,n}$$

$$\text{Expected value}(\mu) = E(X) = ?$$

$$\text{Variance}(\sigma^2) = V(X) = ?$$

Derivation of Expected value

$$E(X) = \sum_{x=0}^{n} x\binom{n}{x} p^x (1-p)^{n-x}$$

Using the identity: $x\binom{n}{x} = n\binom{n-1}{x-1}$

$$E(X) = \sum_{x=1}^{n} n\binom{n-1}{x-1} p^x (1-p)^{n-x}$$

$$E(X) = np \sum_{x=1}^{n} \binom{n-1}{x-1} p^{x-1} (1-p)^{(n-1)-(x-1)}$$

Let $y = x - 1$: $E(X) = np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{(n-1)-y} = np$

$$\boxed{E(X) = np}$$

Derivation of Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Note that: $X^2 = X(X-1) + X$

$$E(X^2) = E[X(X-1)] + E(X)$$

$$E[X(X-1)] = \sum_{x=0}^{n} x(x-1)\binom{n}{x} p^x (1-p)^{n-x}$$

Using the identity: $x(x-1)\binom{n}{x} = n(n-1)\binom{n-2}{x-2}$

$$E[X(X-1)] = n(n-1)p^2$$

$$E(X^2) = n(n-1)p^2 + np$$

$$\text{Var}(X) = n(n-1)p^2 + np - (np)^2$$

$$\boxed{\text{Var}(X) = np(1-p)}$$

Problem 1

A fair coin is tossed 10 times. Find the probability that the number of heads obtained is more than the number of tails.

Problem 2

Each question in an examination has a probability of 0.7 of being answered correctly. A student attempts 20 independent questions. Find the probability that the student answers exactly 14 questions correctly.

Problem 3

A factory produces light bulbs, of which 2% are defective. If 50 bulbs are selected at random, find the probability that at most one bulb is defective.

Problem 4

A basketball player makes a free throw with probability 0.8. If the player attempts 15 free throws, find the probability that the player makes at least 12 successful shots.

Problem 5

An online advertisement converts 10% of viewers into buyers. If the advertisement is shown to 100 viewers, find the expected number of buyers.

Negative Binomial Distribution

Let

- r = desired (fixed) number of successes,
- p = probability of success in each trial,
- X = number of trials required to achieve the r-th success.

Note: The last trial must be a success; therefore, the first $x-1$ trials contain exactly $r-1$ successes.

$$X \sim \mathrm{NB}(r, p)$$

The probability mass function is

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, r+2, \ldots$$

$$\mathbb{E}[X] = \frac{r}{p}; \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

Problem 1

A fair coin is tossed until 3 heads appear. Find the probability that the 3rd head occurs on the 7th toss.

Problem 2

A die is rolled repeatedly until 4 sixes are obtained. Find the probability that the 4th six appears on the 15th roll.

Problem 3

A basketball player makes a shot with probability 0.6. Find the probability that the 5th successful shot occurs on the 9th attempt.

Problem 4

A machine produces defective items with probability 0.08. Find the probability that the 2nd defective item is found on the 10th inspection.

Problem 5

A student answers a multiple-choice question correctly with probability 0.25. Find the probability that the 6th correct answer occurs on the 20th question.

Problem 6

A light bulb works with probability 0.9. Find the probability that the 3rd faulty bulb is found on the 12th test.

Problem 7

A sales agent successfully closes a deal with probability 0.4. Find the probability that the 7th sale happens on the 18th call.

Problem 8

A coin with probability of heads 0.3 is tossed until 5 heads are obtained. Find the probability that the 5th head occurs on the 14th toss.

Geometric Distribution

The geometric distribution is a special case of the negative binomial distribution with

$$r = 1.$$

Let

- p = probability of success in each trial,
- X = number of trials until the first success.

Note: The last trial must be a success.

$$X \sim \mathrm{G}(p)$$

The probability mass function is

$$P(X = x) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots$$

$$\mathbb{E}[X] = \frac{1}{p}; \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}$$

Problem 1

A fair coin is tossed repeatedly until the first head appears. Find the probability that the first head occurs on the 6th toss.

Problem 2

A die is rolled until a six appears. Find the probability that the first six occurs on the 4th roll.

Problem 3

A basketball player makes a shot with probability 0.75. Find the probability that the first successful shot occurs on the 5th attempt.

Problem 4

A machine produces defective items with probability 0.04. Find the probability that the first defective item is found on the 12th inspection.

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Poisson Distribution

Suppose an automatic lathe mass-produces pins.

- Let $n = 20{,}000$ be the total number of pins produced.
- Let $p = \frac{1}{10{,}000}$ be the probability that a pin is perfect.
- Let X denote the number of perfect pins produced.

Since each pin is produced independently with the same probability of being perfect, $X \sim \text{Binomial}(n, p)$.

$$P(X = x) = 20{,}000C_x \left(\frac{1}{10{,}000}\right)^x \left(1 - \frac{1}{10{,}000}\right)^{20{,}000-x}.$$

We observe that

- n is very large,
- p is very small,
- $\lambda = np = 20{,}000 \times \frac{1}{10{,}000} = 2$,

which is neither too large nor too small.

Poisson Distribution

$$X \sim P(\lambda)$$

iff

$$P(X = x) = \frac{e^{-\lambda}\,\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

$$\lambda = \mathbb{E}(X) = np; \qquad \mathrm{Var}(X) = np$$

$$\because \ p \text{ is very small}, \qquad \therefore \ 1 - p \text{ is close to } 1$$

$$\therefore \ np(1-p) \approx np$$

<u>Observe</u>:

We need not know n and p. It is enough if we know their product np, which is denoted as λ.

Problem 1

The average number of customers arriving at a store per hour is 5. Find the probability that exactly 3 customers arrive in an hour.

Problem 2

The average number of accidents at a junction per week is 2. Find the probability that no accident occurs in a week.

Problem 3

A coin is tossed 100 times. If the probability of getting a head on each toss is 0.5, find the probability of obtaining exactly 4 heads.

Problem 4

Emails arrive at a server at an average rate of 6 per minute. Find the probability that exactly 4 emails arrive in a minute.

1. Random Variables
2. Probability distributions
3. Binomial distribution
4. Poisson distribution
5. Problem solving

Problem 1

Find the probability of getting exactly 2 heads in 6 tosses of a fair coin.

Problem 2

Toss a fair coin 100 times and count the number of heads that appear. Find the mean, variance, and standard deviation of this experiment.

Problem 3

If the probability that an individual will suffer a bad reaction from injection of a given serum is 0.001, determine the probability that out of 2000 individuals

1. exactly 3 individuals will suffer a bad reaction,
2. more than 2 individuals will suffer a bad reaction.

Problem 4

Find the probability of getting between 3 and 6 heads inclusive in 10 tosses of a fair coin.

Problem 5

A and B play a game in which their chances of winning are in the ratio 3 : 2. Find A's chance of winning at least 3 games out of the five games played.

Problem 6

An irregular six-faced die is thrown, and the expectation that it will give five even numbers is twice the expectation that it will give four even numbers. How many times in 10,000 sets of 10 throws each would you expect it to give no even number?

Problem 7

The probability of a successful rocket lauching is p. If launching attempts are made until 3 successful launching have occurred

1. What is the probability that exactly 5 attempts will be necessary?
2. What is the probability that fewer than 5 attempts will be necessary? If launching attempts are made until 3 consecutive successful launching occur, What are for the probabilities for
3. exactly 5 attempts,
4. fewer than 5 attempts.

Problem 8

Out of 800 families with 4 children each, how many families would be expected to have

1. 2 boys and 2 girls
2. at least 1 boy
3. atmost 2 girls
4. children of both sexes

Assume equal probabilities for boys and girls.

Problem 9

A communication system consists of n components, each of which will independently function with probability p. The total system will be able to operate effectively if atleast one-half of its components function. For what values of p, 5-component system is more likely to operate effectively than a 3-component system?

Problem 10

Two dice are thrown 120 times. Find the average number of times in which the number of the first dice exceeds the number on the second dice.

Problem 11

The number of monthly breakdowns of a computer is a random variable having a poisson distribution with mean equal to 1.8. Find the probability that this computer will function for a month

1. without a breakdown
2. with only one breakdown
3. with at least one breakdown

Problem 12

It is known that the probability of an item produced by a certain machine will be defective is 0.05. If the produced items are sent to the market in pockets of 20, find the number of pockets containing at least exactly and at most 2 defective items in a consignment of 1000 pocket using

1. Binomial distribution
2. Poisson approximation to binomial distribution

Problem 13

A and B shoot independently until each has hit his own target. The probabilities of their hitting the target at each shot are $3/5$ and $5/7$ respectively. Find the probability that B will require more shots than A.

Thanks!